are sterile. Urine within the urinary bladder is also sterile. In addition to removing microbes by flushing action, urine kills some bacteria due to its low pH and the presence of urea and other metabolic end products (uric acid, hippuric acid, indican, fatty acids, mucin, and enzymes). The acidic environment (pH 3-5) of the vagina is unfavorable to most microbes.

vi) Eye

The conjunctiva is specialized mucus secreting epithelial membrane that lines the interior surface of each eyelid and the exposed surface of the eye ball. It is kept moist by the continuous flushing action of tears. Tears contain large amounts of lysozyme, lactoferrin, and antibody and thus provide chemical as well as physical protection (Figure 13.18).

B. Chemical mediators

- **Antimicrobial peptides**

They are low molecular weight proteins that exhibit broad spectrum antimicrobial activity toward bacteria.

i) Cationic peptides

Cationic peptides are found in humans. There are three generic classes of cationic peptides that have the ability to damage bacterial plasma membrane. Classes of Cationic Peptides are Cathelicidins, Defensins and Histatin

ii) Bacteriocins

Bacteriocins are produced by gram negative and gram positive bacteria. For example, Escherichia coli synthesize bacteriocins called colicins. Colicins causes cell lysis.

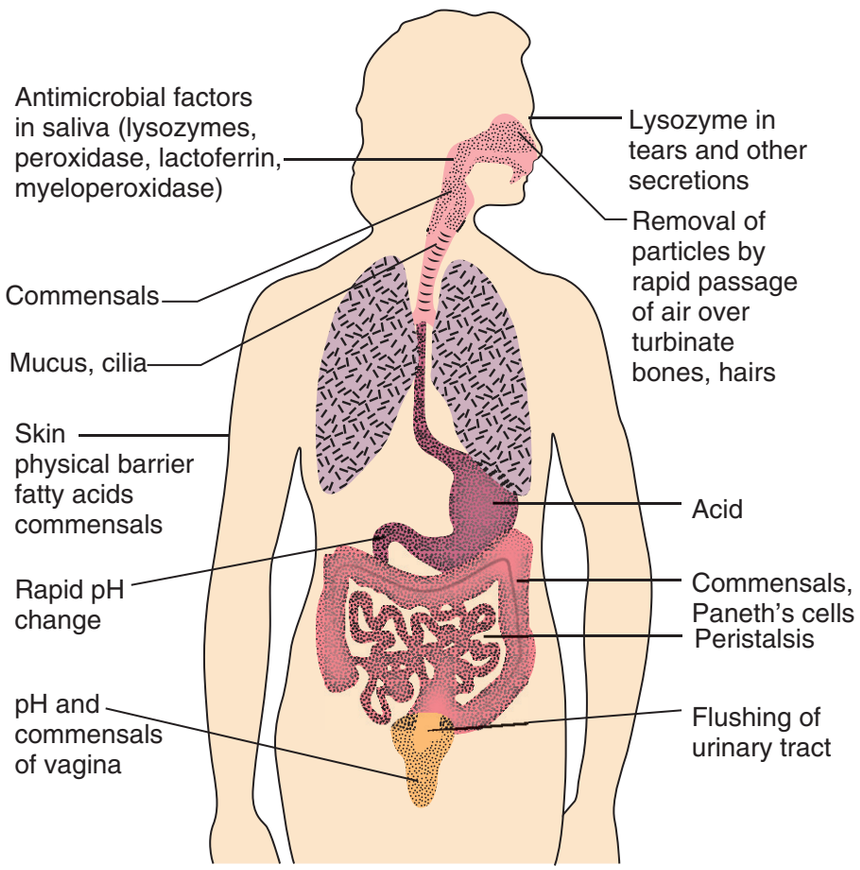


Figure 13.18: Physical Barriers

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